

Write C Program to print "Hello World"

Aim:- To write C Program to print Hello World

Program:-

```
#include <stdio.h>
```

```
void main ( )
```

```
{
```

```
printf ("Hello World\n");
```

```
}
```

Save :- F₂

Compile :- Alt + F9

Execute :- Ctrl + F9

Output :-

Hello World

Write C Program on Arithmetic Operators statically

Aim:- To write C Program on Arithmetic Operators statically

Program:-

```
#include <stdio.h>
#include <conio.h>
void main ()
{
    int a=5, b=10, c;
    clrscr();
    printf("\n... Arithmetic Operations...");
    c=a+b;
    printf("%d + %d = %d", a, b, c);
    c=a-b;
    printf("%d - %d = %d", a, b, c);
    c=a*b;
    printf("%d * %d = %d", a, b, c);
    c=a/b;
    printf("%d / %d = %d", a, b, c);
    c=a%b;
    printf("%d mod %d = %d", a, b, c);
    getch();
}
```

Save :- F2

Compile :- Alt + F9

Execute :- Ctrl + F9

Output :-

.... Arithmetic Operations

$$5 + 10 = 15$$

$$5 - 10 = -5$$

$$5 * 10 = 50$$

$$5 / 10 = 0$$

$$5 \bmod 10 = 5$$

Write a C Program on logical operators dynamically

Aim :- To write C Program on logical operators dynamically

Program :-

```
#include <stdio.h>
#include <conio.h>
void main ( )
{
    int a, b;
    clrscr();
    printf("enter a, b values");
    scanf ("%d %d", &a, &b);
    printf ("\n... logical operators ...");
    printf ("%d & %d = %d \n", a, b, (a & b));
    printf ("%d || %d = %d \n", a, b, (a || b));
    printf ("%d ! %d = %d \n", a, b, (a ! b));
    getch();
}
```

Save :- F2

Compile :- Alt + F9

Execute :- Ctrl + F9

Output :-

... logical operators ...

enter a, b values 10 20

$(10 > 20) \&\& (10 < 20) = 0$

$(10 > 20) \|\ (10 < 20) = 1$

~~$(10 > 20) \&\& (10 < 20) = 0$~~

Write a C-Program for relational operators statically

Aim :- To write C-Program on relational operators

statically
Program :-

```
#include <stdio.h>
```

```
void main ( )
```

```
{
```

```
int a=10, b=12;
```

```
clrscr ( );
```

```
printf (". . . . relational operations . . . . \n");
```

```
printf ("%d > %d = %d \n", a, b, (a > b));
```

```
printf ("%d < %d = %d \n", a, b, (a < b));
```

```
printf ("%d >= %d = %d \n", a, b, (a >= b));
```

```
printf ("%d <= %d = %d \n", a, b, (a <= b));
```

```
printf ("%d == %d = %d \n", a, b, (a == b));
```

```
printf ("%d != %d = %d \n", a, b, (a != b));
```

```
getch ( );
```

```
}
```

Save :- F₂

Compile :- Alt + F₉

Execute :- ctrl + F₉

Output

... relational operators ...

$$10 > 12 = 0$$

$$10 < 12 = 1$$

$$10 \geq 12 = 0$$

$$10 \leq 12 = 1$$

$$10 == 12 = 0$$

$$10 != 12 = 1$$

Write a C Program on Assignment Operator statically

Aim :- To write C-Program on Assignment Operator statically

Program :-

```
#include <stdio.h>
#include <conio.h>
void main ( )
{
    int a,b,c;
    clrscr ( );
    a=10, b=20;
    printf ( "\n... Assignment Operations ...");
    c = a += b;
    printf ( "%d += %d = %d", a, b, c);
    c = a -= b;
    printf ( "%d -= %d = %d", a, b, c);
    c = a *= b;
    printf ( "%d *= %d = %d", a, b, c);
    c = a /= b;
    printf ( "%d /= %d = %d", a, b, c);
    c = a %= b;
    printf ( "%d mod = %d = %d", a, b, c);
    getch ( );
}
```

Save :- F2

Compile :- Alt + F9

Execute :- Ctrl + F9

Output

... Assignment Operators ...

$$10 + = 20 = 30$$

$$10 - = 20 = 10$$

$$10 * = 20 = 200$$

$$10 / = 20 = 0$$

$$10 \text{ mod } = 20 = 0$$

Write C Program on bitwise operators dynamically

Aim :- To write C Program on bitwise operators

dynamically
Programs :-

```
#include <stdio.h>
#include <conio.h>
void main ()
{
    int a, b, c;
    clrscr ();
    printf ("\n enter a, b values = ");
    scanf ("%d %d", &a, &b);
    printf ("\n .... bitwise operation ... ");
    c = a & b;
    printf ("\n %d & %d = %d", a, b, c);
    c = a | b;
    printf ("\n %d | %d = %d", a, b, c);
    c = a ^ b;
    printf ("\n %d ^ %d = %d", a, b, c);
    c = a >> b;
    printf ("\n %d >> %d = %d", a, b, c);
    c = a << b;
    printf ("\n %d << %d = %d", a, b, c);
    c = ~a;
    printf ("\n ~ %d = %d", a, c);
    getch ();
}
```

Save :- F2

Compile :- Alt+F9 Execute :- Ctrl+F9

Output :-

... bitwise Operators ...

enter a, b values 10 1

$$10 \& 1 = 0$$

$$10 | 1 = 11$$

$$10 \wedge 1 = 11$$

$$10 >> 1 = 5$$

$$10 << 1 = 20$$

$$\sim 10 = 1$$

Write C-Program on conditional operations statically

Aim :- To write C Program on conditional operators statically

Program :-

```
#include <stdio.h>
#include <conio.h>
void main ( )
{
    int a=20, b=30, c;
    printf ("\n... conditional operations....");
    c = a > b ? a : b;
    printf ("%d > %d ? %d : %d = %d", a, b, a, b, c);
    c = a < b ? a : b;
    printf ("%d < %d ? %d : %d = %d", a, b, a, b, c);
    getch ( ) ;
}
```

Save :- F₂

Compile :- Alt + F9

Execute :- Ctrl + F9

Output :-

.... conditional operators

$20 > 30 ? 20 : 30 = 30$

$20 < 30 ? 20 : 30 = 20$

Write C Program to find even or odd number.

Aim:- To write C Program to find even or odd number

Program:-

//C Program to find even or odd number

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main ( )
```

```
{
```

```
int a;
```

```
printf("enter a value = \n");
```

```
scanf("%d", &a);
```

```
b = a % 2;
```

```
(b == 0) ? printf("even") : printf("odd");
```

```
getch();
```

```
}
```

Save :- F2

Compile :- ALT + F9

Execute :- ~~Ctrl~~ + F9

Output:

enter a value = 52
even

enter a value = 57
odd

Write C Program to calculate Simple Interest

Aim :- To write C Program to calculate Simple Interest

Program :-

// C Program to calculate Simple Interest

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main ( )
```

```
{
```

```
float p, t, r, SI;
```

```
printf ("enter principle , numbers of years , rate of  
interest = ");
```

```
scanf ("%f%f%f", &p, &t, &r);
```

```
SI = (p*t*r)/100;
```

```
printf ("\n Simple Interest = %f", SI);
```

```
getch();
```

```
}
```

Save :- F2

Compile :- Alt + F9

Execute :- ctrl + F9

Output :-

enter p, t, r values = 5 5 10

Simple interest = 2.500000

Write C Program to ^{reverse} swap 3 numbers

Aim :- To write C Program to ^{reverse} swap 3 numbers

Program :-

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
int a, n, rev = 0;
```

```
printf("enter number = \n");
```

```
scanf("%d", &n);
```

```
a = n % 10;
```

```
rev = rev * 10 + a;
```

```
n = n / 10;
```

```
a = n % 10;
```

```
rev = rev * 10 + a;
```

```
n = n / 10;
```

```
a = n % 10;
```

```
rev = rev * 10 + a;
```

```
n = n / 10;
```

```
a = n % 10;
```

```
rev = rev * 10 + a;
```

```
printf("reverse of number = %d", rev);
```

```
getch();
```

```
}
```

Save :- F2

compile :- ALT + F9

Execute :- CTRL + F9

Output :-

enter number = 2345

reverse of number = 5432

Write C Program to find biggest of 3 numbers

Aim:- To write C Program to find biggest among 3 numbers using nested if.

Program:-

//C Program to find biggest among 3 numbers

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
int a,b,c;
```

```
printf("enter 3 numbers = \n");
```

```
scanf("%d%d%d", &a,&b,&c);
```

```
if (a > b)
```

```
{
```

```
if (a > c)
```

```
printf("%d is greatest", a);
```

```
else
```

```
printf("%d is greatest", c);
```

```
}
```

```
else
```

```
{
```

```
if (b > c)
```

```
printf("%d is greatest", b);
```

```
else
```

```
printf("%d is greatest", c);
```

```
}
```

```
getch();
```

```
}
```

Save :- F2

Compile :- Alt+F9

Execute :- ctrl+F9

Output:-

enter 3 numbers = 8 6 2

8 is greatest

Write C Program to print days of week using Switch.

Aim:- To write C Program to print days of week using switch.

Program:-

// C Program on days of week

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
int day;
```

```
printf("enter no of the day:");
```

```
scanf("%d", &day);
```

```
Switch(day) {
```

```
case - 1 : printf("\n Monday");
```

```
break;
```

```
case - 2 : printf("\n Tuesday");
```

```
break;
```

```
case - 3 : printf("\n Wednesday");
```

```
break;
```

```
case - 4 : printf("\n Thursday");
```

```
break;
```

```
case - 5 : printf("\n Friday");
```

```
break;
```

```
case - 6 : printf("\n Saturday");
```

```
break;
```

```
case - 7 : printf("\n Sunday\n");
```

```
break;
```

```
default : printf("\n enter numbers b/w 1 and 7");
```

```
break;
```

```
}  
getch();  
}
```

Save :- F2

Compile :- Alt+F9

Execute :- Ctrl+F9

Output :-

enter noob the day = 5

Friday.

Write C Program on arithmetic operators using switch

Aim:- To write C Program on arithmetic operators using switch.

Program:-

```
// C Program on arithmetic operators
#include <stdio.h>
#include <conio.h>
void main()
{
    int a, b;
    char c;
    printf("Enter two numbers:");
    scanf("%d %d", &a, &b);
    printf("Enter any arithmetic operator +, -, *, /, %");
    scanf("%c", &c);
    switch(c) {
        case '+': printf("Sum = %d", a + b);
                    break;
        case '-': printf("Difference = %d", a - b);
                    break;
        case '*': printf("Product = %d", a * b);
                    break;
        case '/': printf("Quotient = %d", a / b);
                    break;
        case '%': printf("Mod = %d", a % b);
                    break;
        default: printf("Invalid operator");
                    break;
    }
    getch();
}
```

Save :- F_2

Compile :- $Alt + F_9$

Execute :- $Ctrl + F_9$

Output :-

enter two numbers : 6 . 2

enter any arithmetic operator +, -, *, /, % : /

quotient = 3

Write C Program to print n natural numbers

Aim:- To write C Program to print n natural numbers using while loop

Program:-

// C Program to print n natural numbers

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
int i=1, n;
```

```
printf("enter n value= \n");
```

```
scanf("%d", &n);
```

```
while (i <= n)
```

```
{
```

```
printf("%d \n", i);
```

```
i++;
```

```
}
```

```
getch();
```

```
}
```

Save :- F2

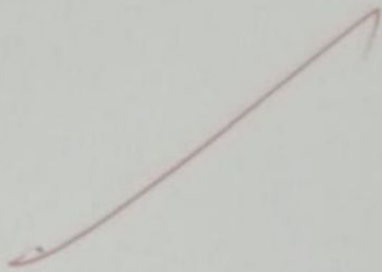
Compile :- Alt+F9

Execute :- Ctrl+F9

Output:-

enter n value = 6

1
2
3
4
5
6



Write C Program to check entered year is leap year or not using goto.

Aim:- To write C Program to check whether entered year is leap year or not

Program:-

// C Program on checking leap year or not

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
int year;
```

```
printf("Enter year:");
```

```
scanf("%d", &year);
```

```
if((year % 4 == 0) && (year % 100 != 0))
```

```
goto leap;
```

```
else
```

```
goto no leap;
```

```
leap;
```

```
printf("%d is leap year", year); return;
```

```
no leap;
```

```
printf("%d is not leap year", year);
```

```
getch();
```

```
}
```

Save :- f2

Compile :- Alt + F9

Execute :- Ctrl + F9

Output :-

enter year : 2020

2020 is leap year

Write C Program to check for armstrong number

Aim:- To write C Program to print if a number is armstrong

Program:-

//C Program to check for armstrong number

```
#include <stdio.h>
#include <conio.h>
void main()
{
    int n, a, sum = 0, m;
    printf("enter n value = \n");
    scanf("%d", &n);
    m = n;
    while (n != 0)
    {
        a = n % 10;
        sum = sum + (a * a * a);
        n = n / 10;
    }
    if (sum == m)
        printf("%d is armstrong", m);
    else
        printf("%d is not armstrong", m);
    getch();
}
```

Save :- F2

Compile :- Alt + F9

Execute :- Ctrl + F9

Output:-

enter n value = 370

370 is armstrong

Write C Program to print multiplication table 10x10

Aim :- To write C Program to print multiplication table 10x10

Program :-

// C Program on 10x10 multiplication table

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
    int i, j, num;
```

```
    for (i=1; i<=10; i++)
```

```
    {
```

```
        num = i;
```

```
        for (j=1; j<=10; j++)
```

```
        {
```

```
            printf("%3d\t", (num * j));
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    getch();
```

```
}
```

Save :- f2

Compile :- ALT + F9

Execute :- CTRL + F9

Output :-

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40
5	10	15	20	25	30	35	40	45	50
6	12	18	24	30	36	42	48	54	60
7	14	21	28	35	42	49	56	63	70
8	16	24	32	40	48	56	64	72	80
9	18	27	36	45	54	63	72	81	90
10	20	30	40	50	60	70	80	90	100

Write C Program to print the pattern

```

*
* *
* * *
* * * *

```

Aim :- To write C Program to print given pattern

Program :-

// C Program on given pattern

```
#include <stdio.h>
```

```
void main()
```

```
{
```

```
int row, i, j;
```

```
printf("enter no of rows : ");
```

```
scanf("%d", &row);
```

```
for (i = 1; i <= row; i++)
```

```
{ for (j = 1; j <= i; j++)
```

```
{
```

```
printf("* ");
```

```
}
```

```
printf("\n");
```

```
}
```

```
getch();
```

```
}
```

Save :- F₂

Compile :- Alt+F8

Execute :- Ctrl+F9